

SECURITY AND TECH

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In the digital modern age, technology has become heavily combined into basically every part of human life. From communication and banking to healthcare and national defense, these technological systems are key for everyday functions. Although, this rise of reliance on technology has additionally brought in risks, specifically in the form of cyber threats. Cybersecurity has grown to be one of the most crucial components of everyday technological infrastructure. And as cyber threats continue to become more refined, guaranteeing the security of technology is not an option but more as a need for individuals, organizations, and governments. In addition to everyday utility, technology has also changed how societies operate on a global scale. Businesses now perform across borders digitally, governments collect and store immense databases of citizen information and users rely on online platforms for everyday life such as education, work, and even social interaction. This interconnected environment raises convenience but also increases the attack surface for these cybercriminals. The outcome of this results in even a single vulnerability can bring about a vast amount of consequences, which will affect millions of users at the same time. This causes cybersecurity to become not just a protective measure, but a requirement in order to maintain modern life.

Cybersecurity is crucial because it protects private information and assures the proper role of digital systems. Because of the increase of cloud computing, internet, and interconnected devices, various amounts of data are created and kept online. This includes sensitive personal data, financial records, intellectual property, etc., which are all very valuable targets for cybercriminals. Based on research, cybersecurity revolves around technologies and processes that are made to defend computers, networks, and data from access and attacks from the unauthorized. Without these proper security practices, individuals as well as organizations are at risk to data breaches, identity theft, and financial losses. The global economy relies heavily on having a protected digital infrastructure. Cyberattacks can interfere with businesses and cause financial damage and hurt reputations. Cybercrime has been calculated to charge the global economy hundreds of billions of dollars annually. This highlights that cybersecurity isn't just a technical issue but is also known for its societal and economic concern. Additionally, the government also depends on cybersecurity to defend national security. Essential infrastructure like transportation systems, power grids, and communication networks are digitally rising, which allows them to become possible targets for cyber warfare. Which stresses the importance of strong cybersecurity practices to protect national stability and safety. Moreover, the need for cybersecurity broadens to personal privacy as well as individual rights. In a world where sensitive data is continuously being stored from social media activity, to online purchases, keeping that kind of information protected is important to keep trust with users. If the person feels like their data is not protected, they might be less entwined to engage with that digital service, which will slow down technological progress. Therefore, cybersecurity plays a very important role in keeping confidence in digital advancements and supporting continued growth in technology.

As technology continues to grow, so do the methods that cybercriminals create to hack it. There are numerous common types of cyber threats that people and organizations deal with. Malware applies to malicious software that is made to destroy or disrupt systems. This includes worms, ransomware, viruses, and spyware. Ransomware attacks are becoming a much more common attack which is when attackers steal data and want payment for its release. Phishing entails fooling people into giving out sensitive information by pretending to be a trustworthy source. These attacks are commonly seen through emails or fake websites. DoS attacks are made to overwhelm systems and become unusable to users. This can interfere with business and services which can ultimately lead to financial losses. Data breaches happen when an unauthorized user has access to sensitive data. This can cause financial fraud, loss of trust, or even identity theft. Not all threats belong to external attackers. Users or employees who have access to systems can always abuse their privileges regardless of if it is intentional or accidental. And because of these threats, it spotlights the importance for strong cybersecurity strategies to spot, prevent, and respond to the attacks. Another growing type of threats include social engineering attacks, in which cybercriminals manipulate users into giving up sensitive information through psychological tricks instead of technical methods. These kinds of attacks are specifically dangerous because they use human behavior instead of system vulnerabilities. Additionally, advanced persistent threats (APTs) are longer term attacks that are usually performed by extremely high skilled groups that could sometimes even be state-sponsored actors. These attacks aim to continue to be unseeable for longer periods of time while taking valuable information. The growth in complexity of these threats really push the need for both technical protection and user awareness.

Growing technologies play an important role in developing and challenging cybersecurity. Technologies like AI, machine learning, and cloud computing have really changed how security systems work. AI and ML are growingly utilized in cybersecurity to spot threats and react to attacks in real time. These technologies can calculate large pools of data and detect patterns that might symbolize malicious activity. An example of this is how AI can spot abnormal network behavior and notify security teams before the attack can happen. Although, AI creates additional challenges. Cybercriminals can also use AI to create more complicated attacks which makes it more difficult for traditional security systems to spot them. It starts a cycle where both defenders and attackers use technology to be one step ahead. Cloud computing allows organizations to hold and access data over the internet instead of using local servers. While this gives more flexibility, it can also bring about security risks. Data stored in the cloud makes the information vulnerable to breaches if it isn't secured properly. As spotlighted in recent studies, the spread of digital services and cloud technologies has significantly grown in cyber threats which pushes the need for better security measures. The Internet of Things pertains to interconnected devices like smart home appliances, wearable devices, industrial sensors. While IoT strengthens convenience and efficiency, it also raises the number of possible exposed points for cyberattacks. Numerous IoT devices lack strong security measures which leaves them vulnerable for hackers. Additionally, blockchain technology has also grown as a possible tool for improving

cybersecurity. Blockchain offers a decentralized and tamper-protectant way of collecting and storing data, which makes it much tougher for attackers to change information. This kind of technology is already being utilized in financial systems and could very much grow into other areas like identity verification and secure data sharing. Although, like all technologies, blockchain is not fully immune to vulnerabilities and must be utilized with caution.

Despite technological advancements, cybersecurity still deals with multiple challenges. Technology is growing rapidly and security measures can catch up. New devices, platforms, and applications are continuously being created which generates new vulnerabilities. There is a global shortage of people who specialize in cybersecurity which makes it tougher for organizations to properly protect their systems. Modern IT systems are far complex and have numerous interconnected components. And this complexity makes it difficult to identify and fix those vulnerabilities. Many security breaches happen because of human error. Examples of this are things like weak passwords or falling for phishing scams. Even the most advanced protection systems can become vulnerable due to the tiniest mistakes. While technology evolves, so do cyber threats. Attackers are becoming more and more precise, using more complicated techniques to get through those security measures. These issues really push the need for constant improvement in cybersecurity practices and technologies. Another important challenge is the lack of global cybersecurity standards and cooperation. Cyber threats typically cross international borders which makes it difficult for any single country to efficiently manage by themselves. The differences in laws, regulations, and enforcement capabilities can build holes that cybercriminals will use. Therefore, working together internationally and creating standardized policies are very crucial in addressing global cybersecurity risks.

In order to navigate these challenges, there are numerous strategies that can be introduced. Organizations should build and enforce security policies that define the best practices for securing data and systems. Employees should be more educated on the risks and best practices of cybersecurity to lower the possibility of human error. Using more advanced tools like firewalls, encryption, and intrusion detection systems can help defend against cyber threats. Updating software and systems so that they are up-to-date is key for fixing vulnerabilities and stopping attacks. Multi-Factor Authentication (MFA) introduces an additional layer of security by needing several forms of verification in order to permit access to sensitive information. Organizations should have a set plan on how to respond to cyber incidents fast and efficiently. And by introducing these strategies, organizations can really lower the risk and strengthen their general security posture. Organizations are rapidly using a “zero trust” security model. This idea assumes that no user or system should be automatically trusted, regardless of if they are inside the network. This means every access request has to be verified in order to gain access. This significantly lowers the risk of insider threats and unauthorized access. Regular security reviews and penetration testing are also very key plans to help detect and identify weaknesses before attackers even get the chance to exploit them.

The future of cybersecurity will be built by ongoing technological advancements. As digital evolution progresses, so does the importance and need for cybersecurity. An obvious trend is the combination of cybersecurity into every aspect of technological development. Security is not an afterthought anymore, it is a very crucial component of system design. Additionally, the government and organizations are venturing more into cybersecurity research and development to remain ahead of threats as they continue to grow. Another important trend is the utilization of automation in cybersecurity. Automated systems can react to threats much more rapid than human operators which lowers the impact of said attacks. Although, as technology grows and evolves, there will be an increase in challenges as well. The growing use of AI, quantum computing, and other advanced technologies will need new ways to approach security. Quantum computing specifically has the opportunity to really revolutionize cybersecurity. While it can significantly strengthen data processing abilities, it also serves a threat to the encryption methods today. Numerous encryption systems depend on mathematical problems that are much tougher for standard computers to be able to solve, however quantum computers might be able to break them much more effectively. This has led to the building of “quantum-resistant” encryption methods, which aim to arrange for this future challenge.

To conclude, security is a crucial part of modern technology. As society becomes more dependent on these digital systems, the necessity for these strong cybersecurity practices continue to evolve. Cyber threats show very significant risks to not just individuals, but organizations as well and even governments, making cybersecurity key for protecting sensitive data, maintaining trust, and ensuring the firmness of digital infrastructure. While growing technologies will pose new opportunities for strengthening security, they have also brought about new challenges. Addressing these challenges will require an integration of advanced technologies, skilled professionals, as well as effective strategies. Finally, cybersecurity is not only a technical problem but is also a joint responsibility. By highlighting security and remaining informed on these evergrowing threats, society will be able to harness the benefits of technology while decreasing its risks. The success of cybersecurity efforts rely on constant adaption and awareness. As threats continue to evolve, so much the strategies used to fight them. Education, innovation, and collaboration will be key in developing a secure digital future. By taking precautionary steps today, both users and organizations can properly prepare for the challenges of tomorrow to ensure that technology continues to remain a safe and useful tool for all people. There are many opportunities for security to grow, whether it be through the use of AI or having more softwares that can combat these cyberthreats. Society must continue to grow in security as technology continues to grow.

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